Reproducing a CFD-DEM study of a bubbling fluidized bed in Python

A guided breakdown of what we actually did: what we built, how it works, what it reproduced, what it did not, and why

EN 649 (Particulate Flow) · Kaustubh Ashok, 24B1522 · Instructor: Prof. Manaswita Bose, IIT Bombay · Companion to the breakdown of Liu & van Wachem (2019), Powder Technology 343, 145–158 · Code: github.com/kaustubh2005ashok-tech/en649-fluidized-bed

The project at a glance

Question	How much of Liu & van Wachem's CFD-DEM results can a DEM code written from scratch in Python reproduce if the gas is treated in one dimension instead of being solved as a 3-D flow?
System	The NETL SSCP particles (3.256 mm, 1131 kg/m³, Geldart D) in a 0.23 m wide, one-particle-thick (2-D monolayer) bed: 3,380 particles, settled height 0.146 m, bed weight 905 Pa per unit area.
Method	Soft-sphere DEM (paper's contact parameters) in Python + Numba, Gidaspow drag on a 24 × 56 grid of 3 d_p cells. Two gas models: *one-way*, $u_g = U_g/\varepsilon_g$, and *quasi-two-way*, $u_g = (U_g - q_s)/\varepsilon_g$. No gas momentum equation.
Runs	28 velocity ramps (14 beds × 2 couplings, plus 70 more ramps testing update ratio and voidage interpolation) and 8 constant-velocity holds of 15 s at 1, 2, 3 and 4 u_{mf}.
Reproduced	$u_{mf} = 0.990 \pm 0.009$ m/s (paper 1.00, experiment 1.05); plateau pressure drop equal to the bed weight; mean ΔP at 1 and 2 u_{mf}; one-way pressure-fluctuation frequency in the paper's 2–5 Hz band.
Not reproduced	Overshoot and hysteresis in the ΔP–U_g curve; ΔP fluctuation size; discrete bubbles at 2 u_{mf}; any valid state at 3 and 4 u_{mf}, where the bed lifts as a plug and ΔP reaches 6× and 12× the bed weight.
Main message	Integral quantities near u_{mf} follow from a force balance and come out right even with a 1-D gas. Anything that needs gas to flow *around* particles (bubbles, bypass, fluctuations) needs the gas momentum equation, and adding the solids flux to the gas velocity does not supply it.

1. What we set out to do

The reference paper runs a full CFD-DEM model of the NETL Small Scale Challenge Problem: every particle is tracked with Newton's laws, and the gas is solved on a grid with volume-averaged Navier–Stokes equations, two-way coupled through drag. It then scores the model against experiment on pressure drop, minimum fluidization velocity, bubbles and particle velocities.

Our aim was narrower. The proposal set out to rebuild the particle side of that model from scratch, in Python, with the paper's particle and contact parameters, and to replace the gas solver with the simplest treatment that can still fluidize a bed: a prescribed superficial velocity turned into a local gas velocity through the voidage. The point was to see how far that gets, to measure the run-to-run scatter, and to find exactly where it stops working. Read this way, the project asks three questions:

- Can a from-scratch DEM code with a 1-D gas reproduce the paper's minimum fluidization velocity and pressure drop?
- Does letting the gas respond to the solids flux (quasi-two-way coupling) change the answer?
- Which of the paper's results need the gas momentum equation, and how do they fail without it?

2. The system we simulated

The paper's bed is pseudo-2-D: 0.23 m wide and 0.075 m (23 d_p) deep. Ours keeps the width and the particles but is truly two-dimensional: one layer of spheres, one diameter thick, with no front or back walls. All material and contact

values come from the paper's Table 1 through a single file, `params.py`.

Property	Paper (3-D)	This work (2-D monolayer)	Comment
d_p, ρ_p	3.256 mm, 1131 kg/m³	same	
ρ_g, μ_g	1.21 kg/m³, 1.8×10^{-5} Pa·s	same	
k_n, k_t, e_n, μ	3654, 1044 N/m, 0.84, 0.35	same	$k_t/k_n = 2/7$
Bed size	0.23 × 0.075 × 1.22 m	0.23 m × 0.55 m × 1 d_p	hard ceiling at 0.55 m
Particles	92,547	3,380 (52 rows × 65)	settled by gravity from a lattice
Settled height, ε_{mf}	0.167 m, 0.42	0.146 m, 0.432–0.452 (bed 7: 0.441)	2-D packs looser
Bed weight / area	1075 Pa	905 Pa	shorter bed, see box
Fluid grid	22 × 7 × 114, ≈ 3.2 d_p	24 × 56, 9.6 × 9.8 mm ≈ 3 d_p	
Time steps	DEM 20 µs, gas 100 µs	DEM 3.33 µs; voidage every 5 steps	see §3.5
Gas phase	VANS equations, 3-D	no gas equations; 1-D velocity rule	the main difference

> **Consistency checks (own calculation).** Bed mass N m = 3380 × 2.044 × 10^{-5} kg = 0.0691 kg. Weight per monolayer cross-section N m g / (W d_p) = 905 Pa, which is also $(1-\varepsilon)\rho_p$ g h = 0.559 × 1131 × 9.81 × 0.146 m. The paper's 1075 Pa comes from a taller (0.167 m), denser (ε = 0.42) bed, so ΔP is compared as a fraction of each bed's own weight. Ar = 1.43 × 10^6 (Geldart D). Ergun at our measured $\varepsilon_{mf} = 0.441$ gives $u_{mf} = 1.104$ m/s; Wen–Yu gives 0.96 m/s.

3. How we did it

3.1 Code structure

Everything is plain Python with the inner loops compiled by Numba (`@njit`). One DEM step costs about 2.5 ms for 3,380 particles, so a 15 s run is 4.5 million steps and takes about 3 hours.

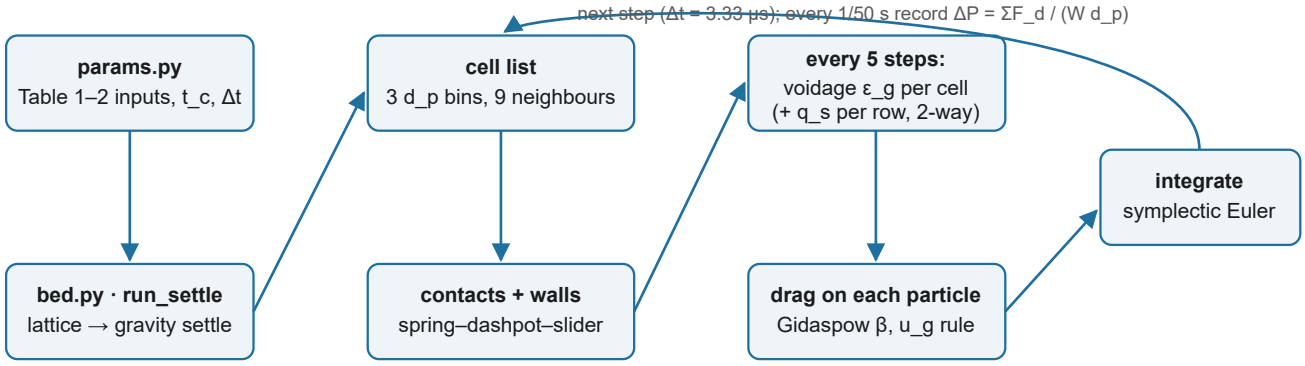


Figure 1. One DEM step as implemented in `sim.step()`. Unlike the paper's loop there is no gas solve: the voidage field (and, for quasi-two-way, the row solids flux) is the only information passed from particles to "gas", and nothing is passed back except drag.

3.2 Particle phase

Each particle obeys Newton's second law with gravity, contact forces from neighbours and walls, and drag:

$$m\frac{d\mathbf{v}_a}{dt} = m\mathbf{g} + \sum_b (\mathbf{F}_{n,ab} + \mathbf{F}_{t,ab}) + \mathbf{F}_{d,a}, \qquad \mathbf{F}_d = \frac{\beta\,(u_g - v_{p,y})\,V_p}{\varepsilon_s}\,\hat{\mathbf{y}}$$

Compared with the paper's Eq. (3), two terms are missing. There is no separate pressure-gradient force $-V_p\nabla p$, because the Ergun branch of β is used in its total-force form (it already contains the support the paper splits between

drag and pressure). And there is no rotation: the tangential force is a Coulomb-capped dashpot with no history spring, and the rotational equation diverged at this stiffness when we tried it, so torque was left out. Drag acts only vertically, since the 1-D gas has no horizontal velocity.

Contacts use the same linear spring–dashpot–slider as the paper. The damping follows from the restitution coefficient, and the time step from the contact time:

$$F_n = k_n\delta - \eta_n v_n \ (\geq 0), \quad \zeta = \frac{-\ln e_n}{\sqrt{\ln^2 e_n + \pi^2}}, \quad \eta_n = 2\zeta\sqrt{k_n m_{\text{eff}}}, \quad t_c = \frac{\pi}{\omega_0\sqrt{1-\zeta^2}}, \quad \Delta t = \frac{t_c}{50}$$

Derived quantity (own calculation)	Value	Comment
m; $m_{eff} = m/2$	2.04×10^{-5} kg; 1.02×10^{-5} kg	same as paper
ζ; η_n	0.055; 0.0214 N·s/m	$\eta_t = \eta_n$
t_c; Δt	166 µs; 3.33 µs	50 steps per contact (paper ≈ 8)
Max overlap at 1 m/s	48 µm = 1.5 % of d_p	soft spring, small overlaps

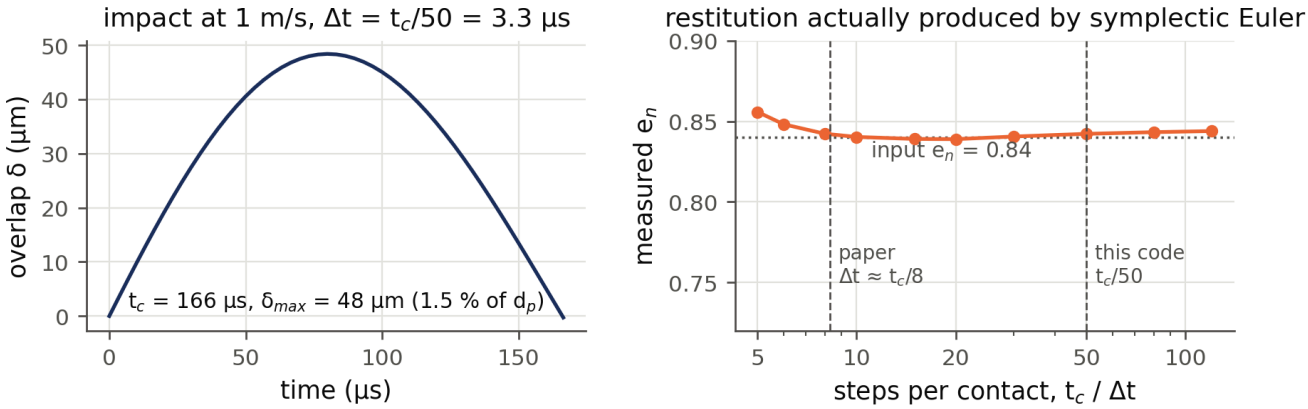


Figure 2. One head-on collision integrated with the code's own scheme (own calculation). Left: overlap history. Right: restitution actually produced at different step sizes. Symplectic Euler returns the input e_n to within 2 % from 5 steps per contact upward, so a single collision does not need $t_c/50$; see §6.

3.3 Gas–solid drag

β is the Gidaspow closure, written for the interstitial slip $u_g - v_p$ so that the pressure gradient is $\beta(u_g - u_s)$:

$$\beta = \begin{cases} \dfrac{150\,\varepsilon_s^2\mu_g}{\varepsilon_g^2 d_p^2} + \dfrac{1.75\,\varepsilon_s\rho_g|u_g - v_p|}{\varepsilon_g d_p}, & \varepsilon_g < 0.8 \text{ (Ergun)} \\ \frac{3}{4}C_D\dfrac{\varepsilon_s\varepsilon_g\rho_g|u_g - v_p|}{d_p}\,\varepsilon_g^{-2.65}, & \varepsilon_g \geq 0.8 \text{ (Wen–Yu)} \end{cases} \qquad C_D = \frac{24}{Re}\left(1 + 0.15Re^{0.687}\right)$$

Substituting the superficial velocity $U = \varepsilon_g u$ into Ergun gives the ε_g^2 and ε_g denominators above. Leaving that factor out was the first bug in the project and made u_{mf} 60 % too high; Checkpoint 2a (§3.6) now tests it. The voidage counts particle volume in each cell as a slab one diameter thick, $\varepsilon_g = 1 - \Sigma V_p/(\Delta x\,\Delta y\,d_p)$, clamped to [0.36, 1], which puts a close-packed monolayer at about 0.46 instead of the areal value of about 0.2 that the correlations cannot handle.

3.4 The two gas models

The paper solves gas continuity and momentum in every cell (its Eqs. 1–2). We solve neither. The gas velocity in a cell is set by a rule:

$$\text{one-way:}\quad u_g = \frac{U_g}{\varepsilon_g} \qquad\qquad \text{quasi-two-way:}\quad u_g = \frac{U_g - q_s}{\varepsilon_g},\quad q_s = \frac{\sum_{\text{row}} V_p v_{p,y}}{W\,\Delta y\,d_p}$$

The quasi-two-way rule is 1-D volume continuity, $\varepsilon_g u_g + \varepsilon_s v_s = U_g$, applied to each row of cells. When solids in a row move up, less gas has to pass through them, and vice versa. With every particle at rest it reduces exactly to the one-way rule (Checkpoint 4a). In both models the bed pressure drop is not solved for but recovered from the total drag, $\Delta P_{bed} = \Sigma F_d/(W\,d_p)$. Neither model lets gas move sideways, and neither has a pressure field that differs across a row.

3.5 Numerical set-up

- **Integration.** Symplectic Euler, $\Delta t = t_c/50 = 3.33$ µs; cell-list neighbour search on the 3 d_p grid.
- **Fluid update.** Voidage (and q_s) refreshed every 5 DEM steps, mimicking the paper's 5 : 1 ratio; drag applied every step from the latest field. 1 : 1 and 10 : 1 were also run.
- **Boundaries.** Side walls and floor use the particle contact law with infinite mass; the top is a hard ceiling at 0.55 m, not a pressure outlet.
- **Beds.** 15 beds (seeds 7, 11–24) built from a jittered lattice and settled under gravity. Bed 23 is excluded from all results because one of its runs failed Checkpoint 3 (plateau at 128 % of the weight); its files are kept.
- **Runs.** Ramps follow the paper: U_g up at 0.2 m/s² to 2.2 m/s, then down, ΔP sampled at 50 Hz. Holds ramp U_g over 2 s (an instant step threw the bed into the ceiling), then hold to 15 s; statistics use 5–15 s, as in the paper. Holds were run on bed 7 only.

3.6 Verification checkpoints

These run in about 30 s (`verify.py`) and were repeated after every change to the force code. Checkpoint 1b alone caught three sign errors during development.

#	Test	Pass criterion	Catches
1a	Binary head-on collision	e_n within 2 % of input	damping, time step
1b	Gravity settling, no gas	KE decays monotonically to < 10^{-5} J	sign errors in any force
2a	β algebra	β·u ≡ Ergun ΔP/L	superficial/interstitial mix-ups
2b	Frozen lattice aligned with cells	$\Sigma F_d/(W\,d_p\,L)$ = Ergun	voidage bookkeeping
3	Fluidized plateau	mean ΔP within 10 % of bed weight	drag summation, mass
4a–d	Volume-flux coupling	at rest ≡ one-way to 10^{-12}; flux and force to 10^{-9}	q_s bookkeeping, slip sign

3.7 How each quantity is measured

Quantity	Procedure
u_{mf}	Descending branch of the ramp only. Fixed-bed part fitted with $aU + bU^2$ (Ergun form), plateau with a constant; u_{mf} is the intersection. Repeated on 14 beds per variant; variants compared with paired t-tests on the same beds.
Pressure drop	$\Sigma F_d/(W\, d_p)$ at 50 Hz; mean and SD over 5–15 s, divided by the bed weight.
Spectrum	Welch PSD of the ΔP signal (256-sample segments, 0.2 Hz resolution); peak frequency.
Flow structure	Voidage field and particle positions at t = 14.5 s, the paper's Fig. 5 time.
Momentum residual	Per cell, the drag that a row-uniform pressure gradient cannot balance (§4.4), time-averaged over 5–15 s.

4. What we found

Quantity	This work vs reference	Verdict
u_{mf}	0.990 ± 0.009 m/s (quasi-two-way), 0.991 ± 0.008 (one-way) vs 1.00 paper, 1.05 experiment	**Good (−1 %, −6 %)**
Plateau ΔP	98.1 % of bed weight in the ramps	**Good**
ΔP–U_g shape	No overshoot (paper ≈ 1130 Pa at 0.8 m/s); ≈ 10 Pa hysteresis vs ≈ 100 Pa; fixed-bed ΔP rises later (0.31 vs ≈ 0.51 of weight at 0.5 m/s)	**Partial**
Coupling effect on u_{mf}	Quasi-two-way − one-way = −0.2 % (95 % CI −0.9 to +0.5 %, p = 0.55)	None detectable
Update ratio 1 : 1 – 10 : 1	All differences < 0.6 %, p ≥ 0.29	None detectable
Mean ΔP at 1, 2 u_{mf}	0.97 and 0.99 × weight vs paper 0.94 × at 2 u_{mf}	**Good**
ΔP fluctuations, 2 u_{mf}	SD 39 / 28 Pa vs swings of 500–1800 Pa in the paper	**Far too small**
Dominant frequency	4.3 Hz one-way (paper 2–5 Hz); 1.2 Hz quasi-two-way	**One-way only**
Flow structure, 2 u_{mf}	Dense base, solids rising at the walls round one large central void; paper has discrete bubbles	**Wrong**
3 and 4 u_{mf}	Bed lifts as a plug to the ceiling; ΔP = 6.2 / 5.8 × and 12.2 / 12.3 × weight vs 0.93 and 0.85 ×	**Invalid state**
Gas momentum balance	Mean residual 0.60 of the bed weight per volume inside the bed, up to 1.5; zero in the paper by construction	**Not conserved**

4.1 Minimum fluidization velocity

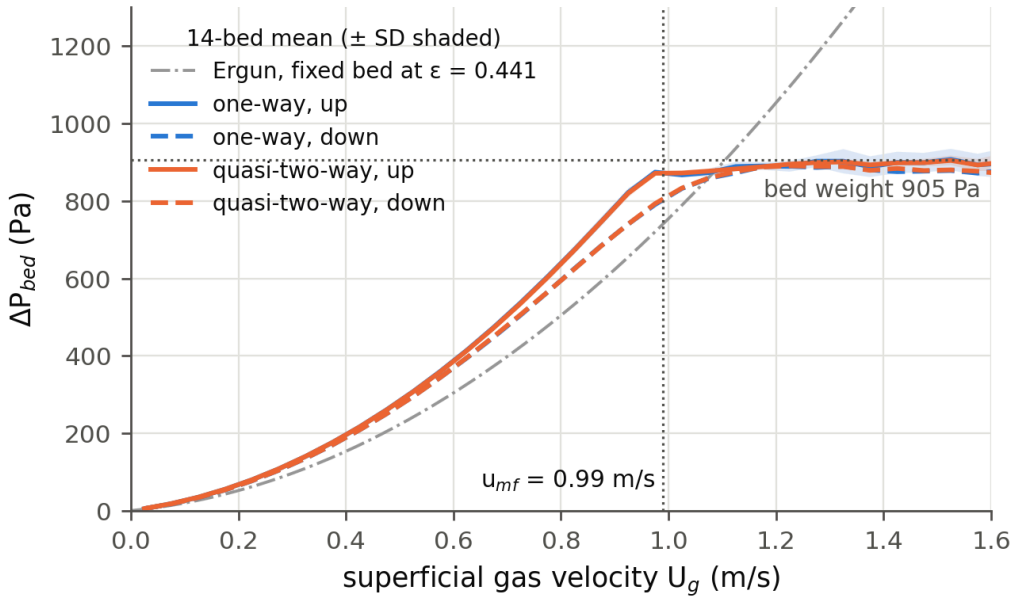


Figure 3. Bed pressure drop through the up and down ramps, mean of 14 beds for each coupling. The two couplings are indistinguishable in the fixed bed because nothing moves there. The grey curve is Ergun for the settled bed at its measured voidage.

The pressure drop rises as the gas squeezes through the packed bed, levels off at the bed weight once the bed is supported, and stays there. Read from the descending branch as in the paper, u_{mf} is 0.990 m/s for the quasi-two-way model and 0.991 m/s for one-way, 1 % below the paper's simulation and 6 % below the experiment. The bed-to-bed standard deviation is about 0.01 m/s, so 14 beds pin the mean to ±0.005 m/s.

Two features of the paper's curve are missing. There is no overshoot on the way up, probably because our bed has no front and back walls for particles to wedge against and no rotation to lock neighbours together. The hysteresis is ten times smaller, which fits a bed with little to unlock. Our u_{mf} also sits 11 % below Ergun at the bed's own voidage, because the simulated curve runs above Ergun before u_{mf}. A likely cause is that drag is summed over cells whose voidage varies around the mean, and β rises steeply in the denser cells; we did not test this.

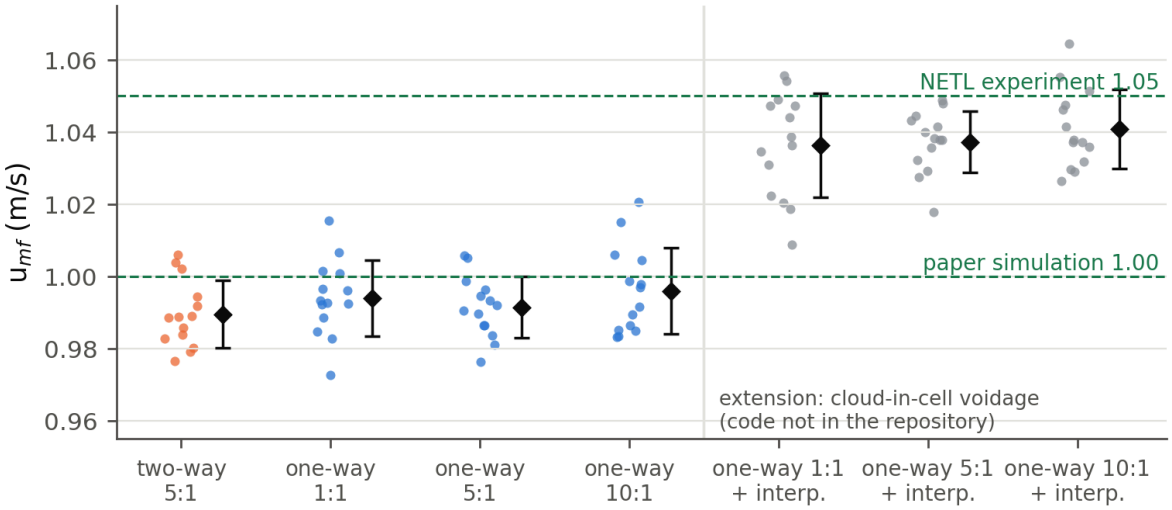


Figure 4. u_{mf} for every bed and variant (dots), with mean ± SD (diamonds). The coupling and the fluid : DEM update ratio change nothing measurable. Interpolating the voidage with cloud-in-cell weights (an extension whose code is not part of the repository) raises u_{mf} by 4–5 %, more than the scatter.

Figure 4 is the clearest result of the project. Neither the coupling nor the update ratio matters for u_{mf}; the way voidage is computed does. That is a warning against reading agreement with experiment as validation: the

interpolated variants land within 1 % of the experiment, but only because one discretisation choice happens to move the answer that way.

4.2 Holding at 1 and 2 u_{mf}

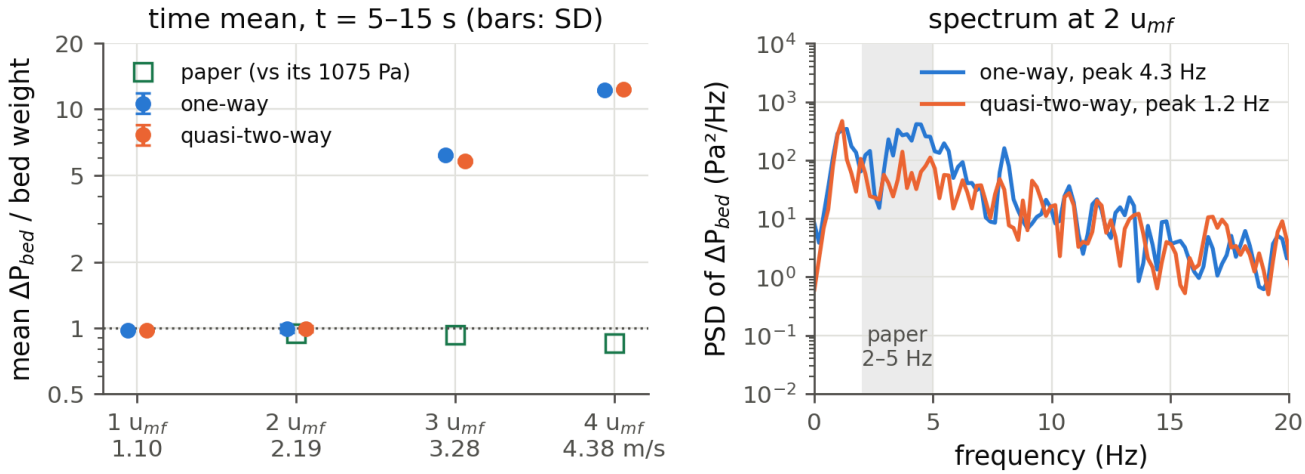


Figure 5. Left: mean pressure drop over 5–15 s as a fraction of the bed weight (log scale; SD bars are smaller than the markers). Right: power spectrum of ΔP at 2 u_{mf} with the paper's dominant band shaded.

At 1 and 2 u_{mf} the mean pressure drop is 0.97 and 0.99 times the bed weight, which is the correct answer for a fully supported bed. The paper's values fall below the weight as velocity rises (0.94, 0.93, 0.85), which its authors tentatively link to gas bypassing the bed in large bubbles, something our model cannot do. The fluctuations, however, are an order of magnitude too small: a standard deviation of 30–40 Pa against swings of hundreds of pascals in the paper. In the paper those swings come from bubbles forming and erupting. The one-way spectrum still peaks at 4.3 Hz, inside the paper's band, so some bed-scale oscillation of the right period exists; in the quasi-two-way run the solids-flux feedback damps it and the peak moves to 1.2 Hz.

4.3 Flow structure, and the plug at 3 and 4 u_{mf}

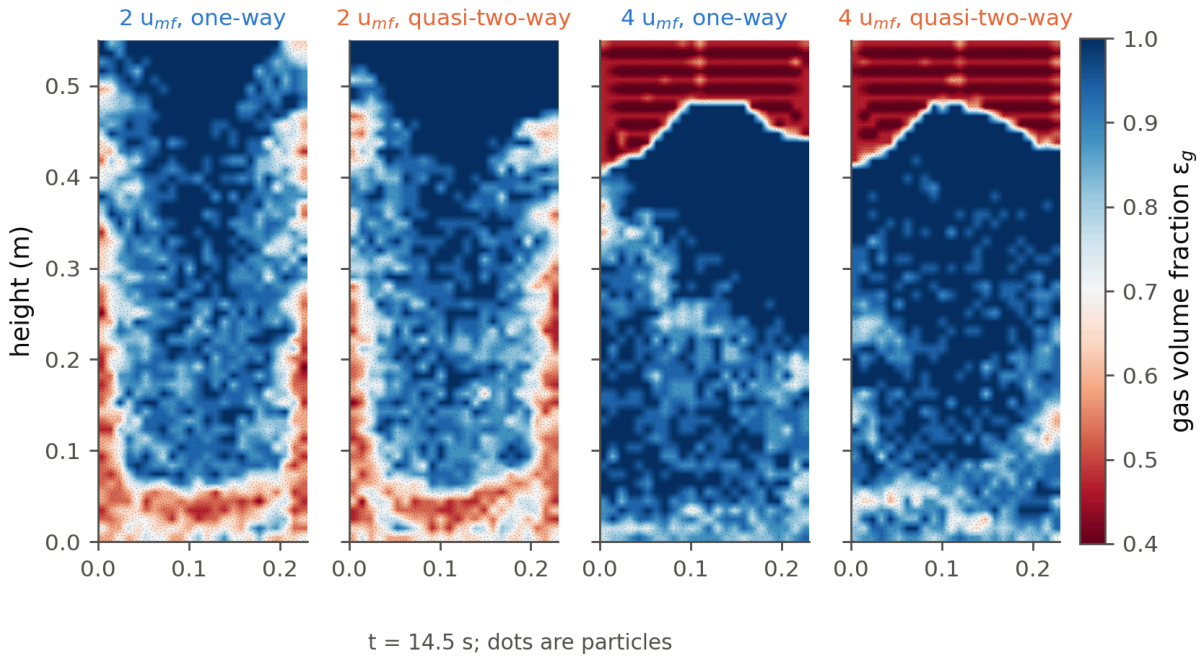


Figure 6. Voidage at t = 14.5 s. At 2 u_{mf} the bed is fluidized but organised as one large central void with solids rising along the walls. At 4 u_{mf} the whole bed is packed against the ceiling (the dense band at the top) with nearly empty space beneath.

At 2 u_{mf} the bed fluidizes: a dense layer sits on the distributor, solids travel up along both walls, and the centre is dilute. The paper shows something different, discrete bubbles that form near the distributor, grow and rise through a dense emulsion. Our bed has no emulsion with bubbles in it. At 3 and 4 u_{mf} it is worse: within two seconds the bed is carried to the top of the domain as a single packed plug and pressed against the ceiling, and the "pressure drop" climbs to 6 and 12 times the bed weight. Both couplings do the same.

> **Check (own calculation).** If all the gas has to pass through the bed packed at ε = 0.44, Ergun over 0.146 m gives 7,160 Pa at 3.28 m/s and 12,570 Pa at 4.38 m/s. The measured 5,240–5,570 and 11,040–11,130 Pa are 73–88 % of that. The gas is being forced through a packed plug, and the ceiling, not gravity, is what holds the plug back.

4.4 Gas momentum is not conserved

The paper's gas momentum equation gives each cell its own pressure gradient, so the drag in every cell is balanced locally. Our models have no pressure field. The most generous reading is a single vertical pressure gradient per row, which spreads the row's drag over its cells in proportion to their gas fraction. The residual is what that leaves unbalanced:

$$R_{\text{cell}} = f_{d,\text{cell}} - \varepsilon_{\text{cell}} \frac{\sum_{\text{row}} f_d}{\sum_{\text{row}} \varepsilon}, \qquad \text{reported as } |R| \,/\, [(1-\varepsilon_{mf})\rho_p g]$$

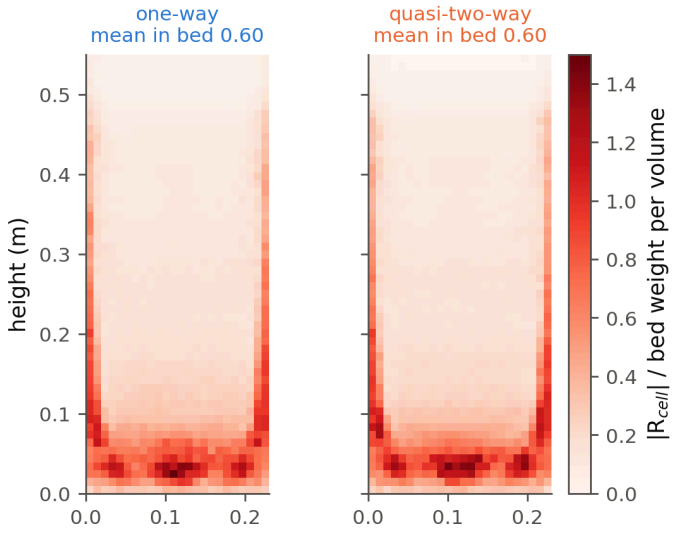


Figure 7. Time-mean residual at 2 u_{mf}. Inside the bed (time-mean $\varepsilon_g < 0.8$) it averages 0.60 of the local bed weight in both couplings, peaking at 1.5 along the walls and at the distributor.

Inside the bed, drag equal to 60 % of the local bed weight is not balanced by any pressure gradient. The largest residuals are in the dense regions: the wall streams and the layer on the distributor. In the paper those regions would raise the local gas pressure and push gas sideways into the dilute centre; that sideways flow is what lets bubbles form and gas bypass the bed. Here it cannot happen, which ties together the failures in §4.2 and §4.3.

5. What it shows

- **Force balance versus flow.** u_{mf}, the plateau, and the mean ΔP at low velocity depend only on the drag on a bed that is just supported, which a 1-D gas captures. Bubbles, fluctuations and bypass depend on how gas is distributed across the bed, which needs the gas momentum equation. The split is sharp: we match the first group to about 1 % and miss the second entirely.
- **Quasi-two-way coupling is not two-way coupling.** Adding the solids flux to the gas velocity conserves gas volume row by row but not momentum, and it changes u_{mf} by −0.2 % (not significant). In a batch bed, solids going up and down in the same row largely cancel in q_s.

- **Scatter needs to be measured.** One bed gives u_{mf} to about ±1 %. Any claim smaller than that needs several beds; with 14 we could show that update ratio and coupling do not matter and that voidage interpolation does.
- **Agreement with experiment is not validation.** The variant closest to experiment (interpolated voidage) is not the more physical one; the 2-D packing and the 1-D gas both shift u_{mf} by similar amounts.

6. Reading it critically

- **2-D monolayer.** No front or back walls, looser packing ($\varepsilon_{mf} \approx 0.44$ vs 0.42), and 905 Pa instead of 1075 Pa. Since $u_{mf} \propto \varepsilon^{3/2}$ in this regime, the looser packing alone moves Ergun's u_{mf} up by about 7 %.
- **No rotation.** Particles slide but cannot roll, which removes one source of interlocking and of energy transfer in collisions. A tangential history spring is needed before rotation can be integrated stably.
- **Time step.** We used $t_c/50$ because symplectic Euler was expected to need it. Figure 2 shows that a single collision is already accurate at $t_c/8$, so the choice is conservative and costs a factor of about six in run time; it was not tested on a full bed.
- **Hard ceiling.** At 3 and 4 u_{mf} the ceiling holds the plug; with a pressure outlet the bed would leave the domain instead. Neither is the paper's behaviour.
- **Single bed for holds.** All 1–4 u_{mf} statistics come from bed 7. The ramps show bed-to-bed scatter of about 1 %, but the fluctuation and spectrum results carry no error bars.
- **Excluded bed.** Bed 23 was removed from all variants to keep comparisons paired, because one of its seven runs failed Checkpoint 3. Including it does not change any conclusion; the cause was not diagnosed.
- **No experimental data.** We compare against the paper's simulation, not against the NETL measurements, so particle velocities and granular temperature were not scored.
- **Gidaspow convention.** Our dense branch uses the standard form; the paper's Table A1 as printed differs in its voidage convention. That changes the dense-branch drag, and so the fixed-bed curve and the u_{mf} read from it; the size of the effect was not measured.

7. The physics behind the failures

7.1 Why a 1-D gas cannot make bubbles

A bubble exists because gas prefers the path of least resistance. Where the bed is locally dilute, the drag per unit gas flow is small, so the pressure field drives more gas there, which pushes particles aside and makes it more dilute still. That instability needs two things: a pressure field that can differ across the bed, and gas that can move sideways to follow it. Our rule fixes the superficial gas velocity in every cell of a row to the same value, so a dilute cell gets no more gas than a dense one, and the feedback never starts. What forms instead is a slower circulation driven by the walls.

7.2 Why the bed becomes a plug

Above u_{mf} the excess gas in a real bed passes through as bubbles, and the emulsion stays near minimum-fluidization conditions (two-phase theory). With no bubbles, all the gas has to go through the particles. At 3 u_{mf} the drag on a uniform bed is then several times its weight, so the whole bed accelerates upward together until something stops it. Our "something" is the ceiling, and the measured ΔP is the force the ceiling needs to apply, which the Ergun check in §4.3 confirms.

7.3 Why the volume-flux coupling barely matters near u_{mf}

Near u_{mf} the particles barely move, so $q_s \approx 0$ and the two rules agree; this is built into the model (Checkpoint 4a). Once fluidized, q_s in a row is the net of solids rising at the walls and falling elsewhere. In a closed batch bed those

nearly cancel, so the correction to u_g is small on average and only changes the dynamics (the 1.2 Hz peak), not the mean.

7.4 Why the update ratio does not matter

Five DEM steps are 16.6 µs. In that time a particle moving at 1 m/s travels 17 µm, about 0.2 % of a fluid cell, so the voidage field barely changes between updates. Even 10 : 1 is far below the time particles take to cross a cell. The paper's ratio comes from a gas solver's stability limit; here it only staggers a bookkeeping step.

8. What would fix it

- **Solve a 2-D gas pressure equation.** A porous-media (Darcy–Ergun) pressure solve per cell, with gas continuity, would give each cell its own ∇p and let gas move sideways. This is the smallest change that could produce bubbles and bypass at 3–4 u_{mf}.
- **Full CFD-DEM cross-check.** Run the same bed in an established CFD-DEM code (Lethe, VANS); five beds have been run so far.
- **Rotation and a tangential spring**, then the ramp overshoot can be tested again.
- **Experimental data.** Obtain the NETL SSCP measurements (Gopalan et al., 2016) to score particle velocities and granular temperature directly.

9. Symbols

Symbol	Meaning	Symbol	Meaning
ε_g, ε_s	gas, solid volume fraction	u_{mf}	minimum fluidization velocity
U_g	superficial gas velocity	u_g, v_p	interstitial gas, particle velocity
q_s	row-averaged solids volume flux	β	inter-phase momentum exchange coefficient
k_n, k_t	normal, tangential stiffness	η_n, ζ	normal damping, damping ratio
e_n	normal restitution coefficient	t_c, Δt	contact duration, DEM time step
W, Δx, Δy	bed width, fluid cell size	R_{cell}	unbalanced drag per cell volume
ΔP_{bed}	bed pressure drop, $\Sigma F_d/(W\, d_p)$	PSD	power spectral density

All figures are original, drawn from this project's run outputs by `presentation/explainer/make_figs.py`. Numbers marked "own calculation" come from `params.py` and the correlations in `drag.py`. Paper values are read from Liu & van Wachem (2019).